

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	BE Computer Science & Engineering	Semester	VI
Subject Code & Name	Machine Learning Laboratory		
Academic Year	2025–2026	Batch	2023–2027

Experiment 8: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

Students must visualize and compare the clusters formed by each algorithm, report results, and analyze performance.

Dataset

Download: Human Activity Recognition Using Smartphones Dataset

- 30 volunteers (aged 19–48). - 6 activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. - Sensors: 3-axis accelerometer and gyroscope (50 Hz). - Preprocessed into windows of 2.56s (128 readings/window) with feature extraction from time and frequency domain.

Graphs and Plots

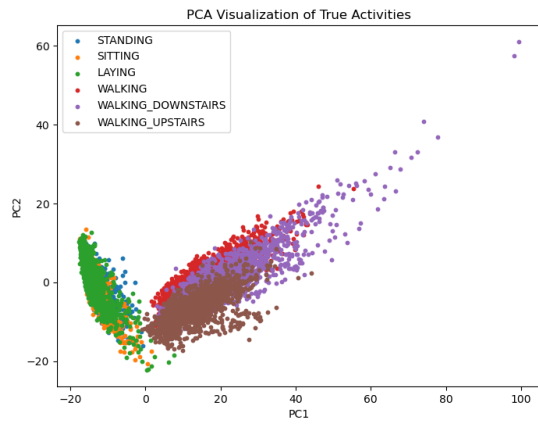


Figure 1: PCA Visualization of True Activities

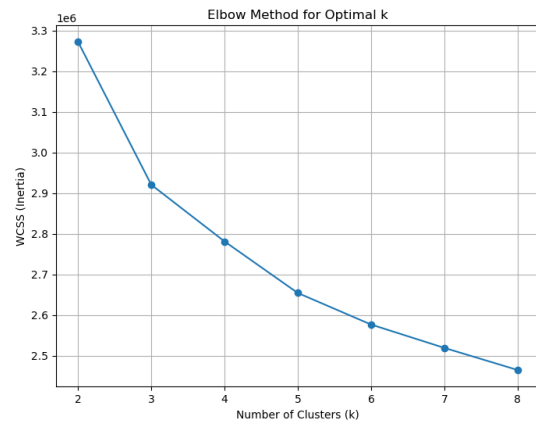


Figure 2: Elbow Plot

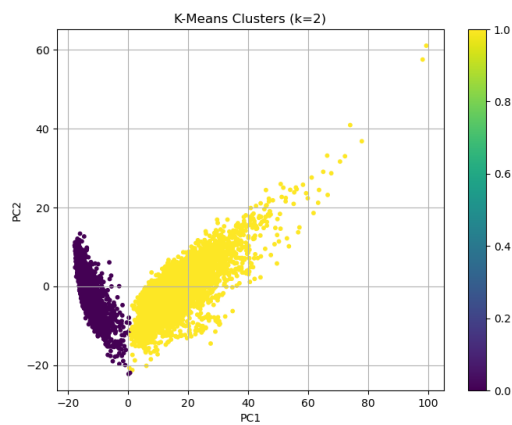


Figure 3: K-means Clusters



Figure 4: True Activity Clusters

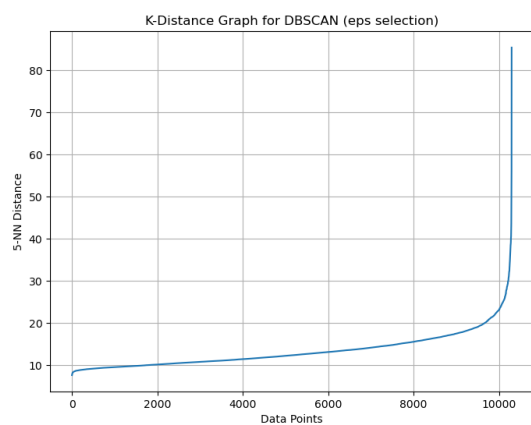


Figure 5: K-Distance Graph for DBSCAN (eps selection)

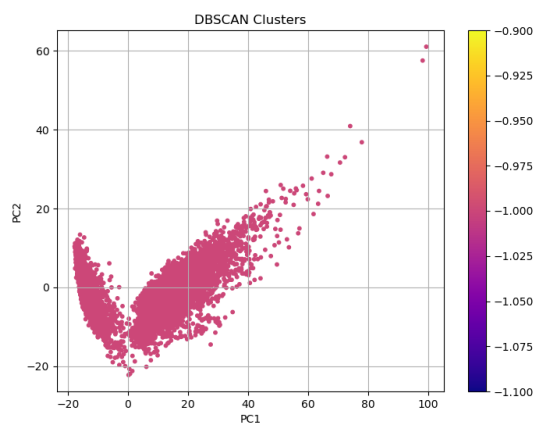


Figure 6: DBSCAN Clusters

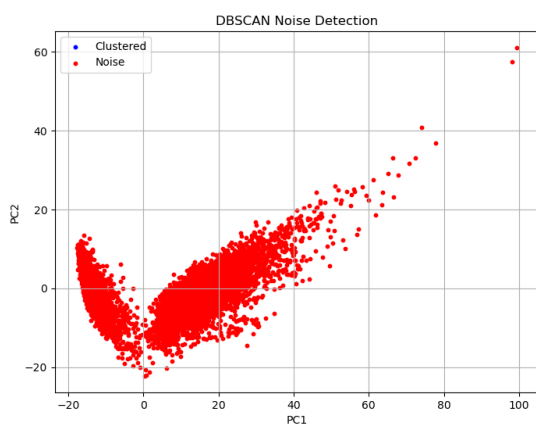


Figure 7: DBSCAN Noise Detection

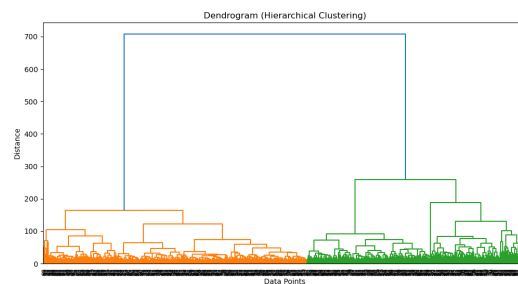


Figure 8: Dendrogram (Hierarchical Clustering)

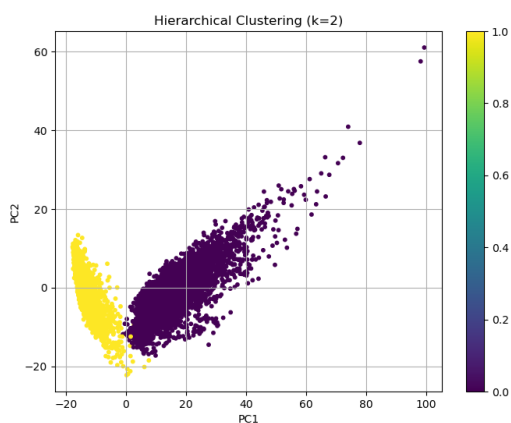


Figure 9: Hierarchical Clustering (k=2)



Figure 10: True Activity Clusters (Reference)

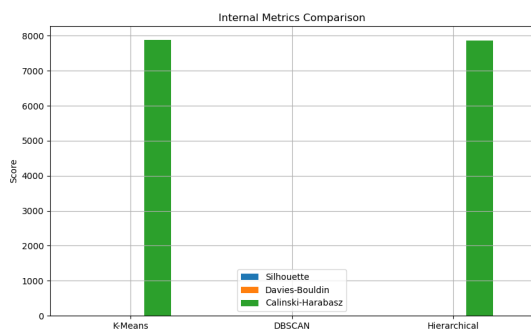


Figure 11: Internal Metrics Comparison

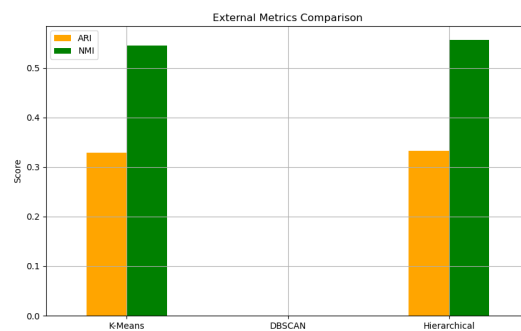


Figure 12: External Metrics Comparison

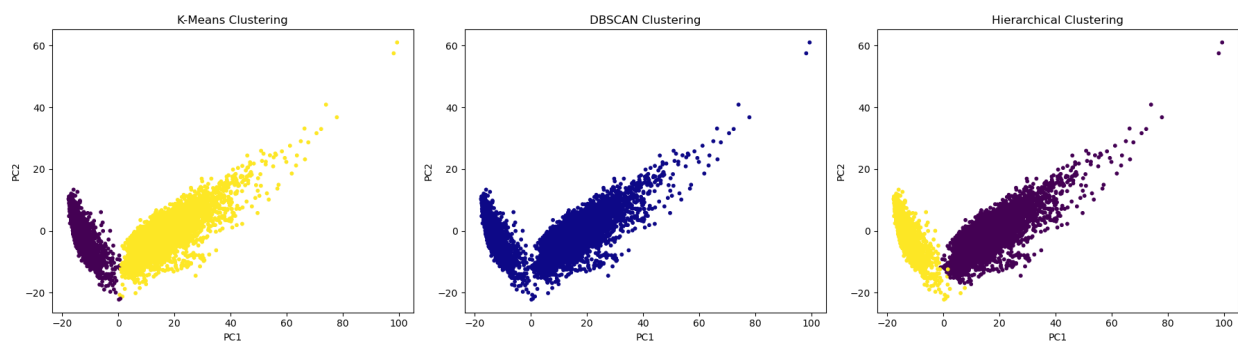


Figure 13: All Clusters Comparison

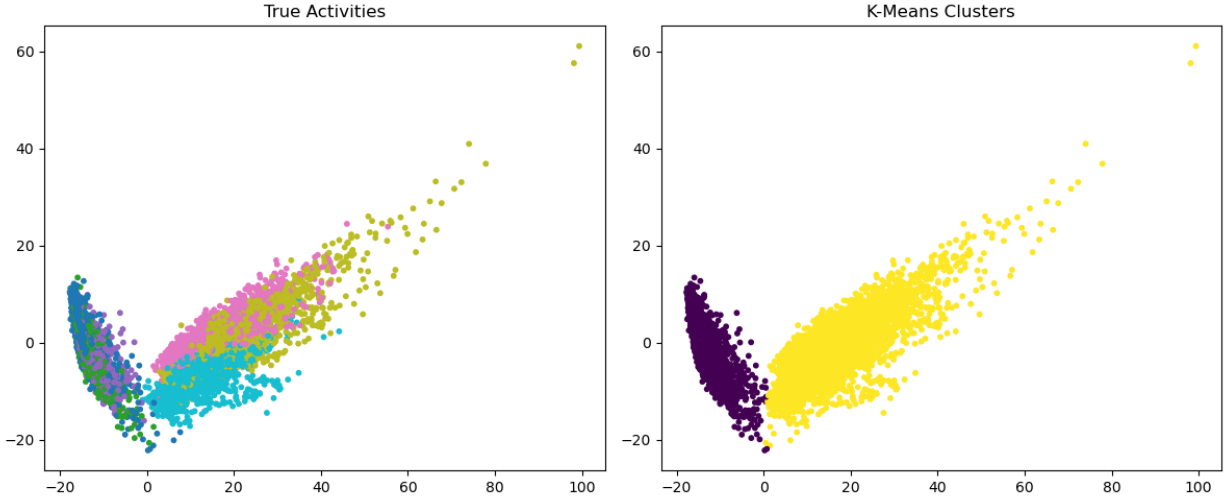


Figure 14: True vs K-means Comparison

Theory

K-Means Clustering

- Objective: Partition data into k clusters by minimizing within-cluster variance. - Distance metric: Euclidean. - Algorithm:

1. Initialize k centroids.
2. Assign points to nearest centroid.
3. Update centroids as mean of assigned points.
4. Repeat until convergence.

Elbow Method for Choosing k

- Compute **Within-Cluster Sum of Squares (WCSS)** for different k values. - Plot k vs. WCSS.
 - The “elbow point” (where reduction in WCSS slows down) suggests the optimal k .

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3.272857e+06	0.393732
3	2.921078e+06	0.315484
4	2.781602e+06	0.150005
5	2.654790e+06	0.128631
6	2.577180e+06	0.109895
7	2.519774e+06	0.081599
8	2.465379e+06	0.074867

DBSCAN

- Density-based clustering algorithm. - Parameters: ϵ (neighborhood radius), *minPts* (minimum points). - Core points, border points, and noise are identified. - Advantage: Can discover clusters of arbitrary shape and handle noise.

Hierarchical Clustering

- Builds hierarchy of clusters using either:

- Agglomerative (bottom-up merging).
- Divisive (top-down splitting).

- Linkage criteria: Single, Complete, Average, Ward's method. - Represented by a dendrogram.

Evaluation Metrics

- **Internal:** Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index.
- **External (w.r.t. true labels):** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI).
- Confusion matrix (optional, by mapping clusters to activities).

Graphs and Visualization

Students should produce:

- Elbow curve: k vs. WCSS.
- Silhouette curve: k vs. Silhouette Score.
- Scatter plots of clusters (after PCA/t-SNE).
- Dendrograms for hierarchical clustering.
- Bar plots of evaluation metrics across algorithms.

Observation Questions

- Which algorithm produced the most meaningful clusters? Why?
- How sensitive was K-Means to the choice of k ?
- Did DBSCAN detect noise or small clusters effectively?
- How does linkage choice (single/complete/ward) affect hierarchical clustering?
- Which internal metric best matched your visual intuition of cluster quality?

Observations and Comparative Analysis

- K-Means clustering achieved the best overall performance, producing compact and well-separated clusters, especially when the optimal value of k was selected using the Elbow and Silhouette methods.
- DBSCAN was effective in identifying noise and outliers; however, its performance was sensitive to parameter tuning and was less effective on high-dimensional data like the HAR dataset.
- Hierarchical clustering (Ward linkage) produced structured and interpretable clusters, but it was computationally expensive and less scalable for large datasets.
- There exists a trade-off between interpretability and performance, where K-Means offers efficiency and accuracy, while DBSCAN and Hierarchical methods provide additional insights such as noise detection and cluster hierarchy.